

R for bioinformatics, Linear and nonlinear regression

HUST Bioinformatics course series

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2025-10-30

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Section 1

section 1: TOC

前情提要

- R basics
- R data wrangler
- R plot
- R string, regular expression
- R parallel computing

本次提要

- linear regression
- nonlinear regression
- modeling and prediction
- **K-fold & X times** cross-validation
- external validation

Section 2

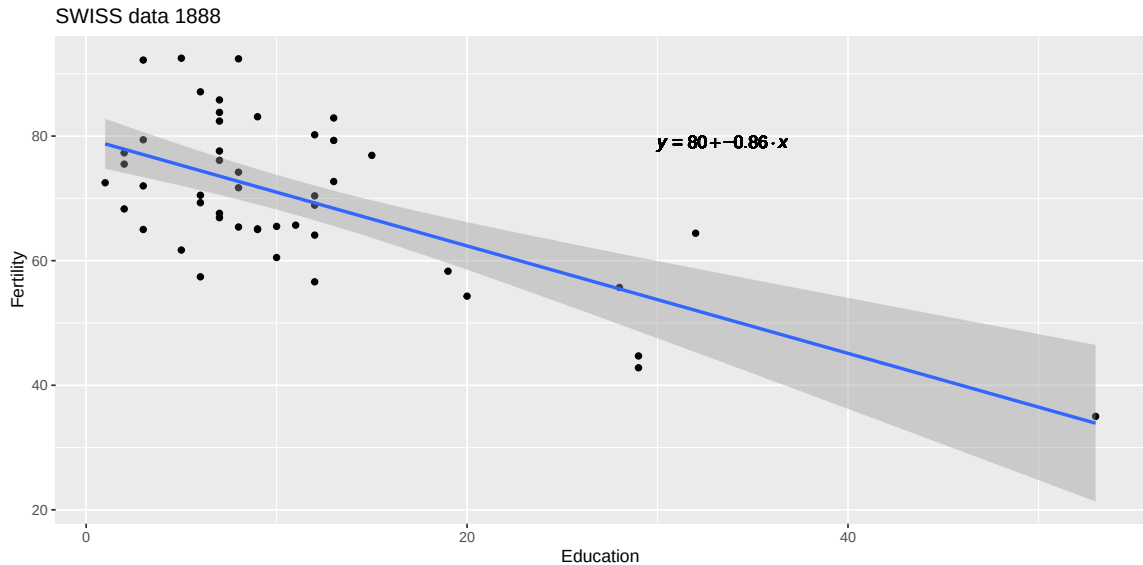
section 2: Linear regression

what is linear regression?

线性回归是利用数理统计中回归分析，来确定两种或两种以上变量间相互依赖的定量关系的一种统计分析方法

- Y 可以被 一个变量 x 解释; 一元线性回归
- Y 可以被 x, z 等多个变量解释; multivariate linear regression

举例



解释

```
m <- lm(Fertility ~ Education, data = swiss);
summary(m);
```

Call:

```
lm(formula = Fertility ~ Education, data = swiss)
```

Residuals:

Min	1Q	Median	3Q	Max
-17.036	-6.711	-1.011	9.526	19.689

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	79.6101	2.1041	37.836	< 2e-16 ***
Education	-0.8624	0.1448	-5.954	3.66e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.446 on 45 degrees of freedom

Multiple R-squared: 0.4406, Adjusted R-squared: 0.4282

F-statistic: 35.45 on 1 and 45 DF, p-value: 3.659e-07



得到 Coefficients

```
coef( m );
```

```
(Intercept)    Education  
  79.6100585   -0.8623503
```

other useful functions

```
# Other useful functions
coefficients(m) # model coefficients
confint(m, level=0.95) # CIs for model parameters
fitted(m) # predicted values
residuals(m) # residuals
anova(m) # anova table
vcov(m) # covariance matrix for model parameters
influence(m) # regression diagnostics
```

R-squared a.k.a R2 是怎么来的?

```
library(magrittr);
library(caret);
predictions <- m %>% predict( swiss ); ## or use fitted(m) instead ...

# Model performance
data.frame(
  RMSE = RMSE(predictions, swiss$Fertility),
  R2 = R2(predictions, swiss$Fertility)
)
```

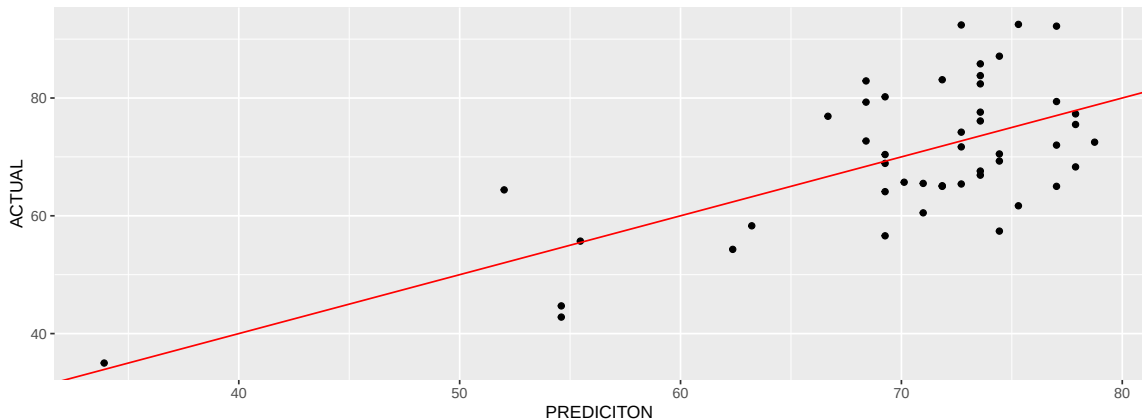
	RMSE	R2
1	9.242865	0.4406156

RMSE mean squared error, the smaller the better **R2** R, higher the better * F-statistic: Higher the better

R-squared a.k.a R^2 是怎么来的? cont.

```
data.frame( PREDICITON = predictions, ACTUAL = swiss$Fertility ) %>%
  ggplot(aes( PREDICITON, ACTUAL )) + geom_point() +
  ggtitle( "SWISS DATA 1888 Fertility" ) +
  geom_abline(intercept = 0, slope = 1, colour = "red")
```

SWISS DATA 1888 Fertility



Multivariate linear modeling

datarium package

```
install.packages("datarium");
```

```
library(datarium);  
head(marketing);
```

	youtube	facebook	newspaper	sales
1	276.12	45.36	83.04	26.52
2	53.40	47.16	54.12	12.48
3	20.64	55.08	83.16	11.16
4	181.80	49.56	70.20	22.20
5	216.96	12.96	70.08	15.48
6	10.44	58.68	90.00	8.64

问题: 广告投放在哪里对销售有帮助??



multivariate linear modeling , cont.

```
m1 <- lm( sales ~ youtube + facebook + newspaper, data = marketing );
m2 <- lm( sales ~ youtube, data = marketing);
m3 <- lm( sales ~ facebook, data = marketing);
summary(m1);
```

Call:
lm(formula = sales ~ youtube + facebook + newspaper, data = marketing)

Residuals:

	Min	1Q	Median	3Q	Max
	-10.5932	-1.0690	0.2902	1.4272	3.3951

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.526667	0.374290	9.422	<2e-16 ***
youtube	0.045765	0.001395	32.809	<2e-16 ***
facebook	0.188530	0.008611	21.893	<2e-16 ***
newspaper	-0.001037	0.005871	-0.177	0.86

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1



facebook vs. youtube

```
coef( m1 );
```

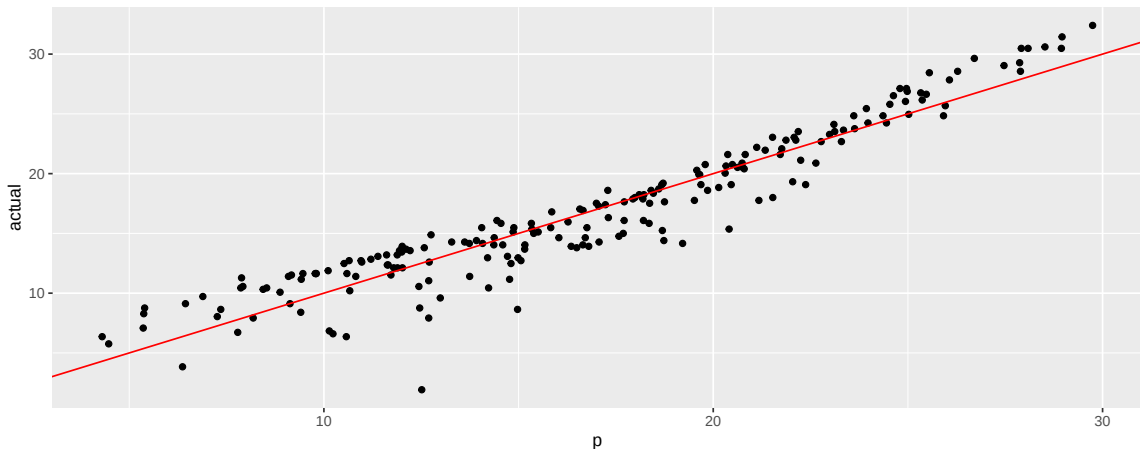
```
(Intercept)      youtube      facebook      newspaper
3.526667243  0.045764645  0.188530017 -0.001037493
```

```
data.frame( YOUTUBE = summary(m2)$r.squared, FACEBOOK = summary(m3)$r.squared);
```

```
      YOUTUBE  FACEBOOK
1 0.6118751 0.3320325
```


predicted vs. actual

```
predicted.m1 <- m1 %>% predict( marketing );
data.frame(p = predicted.m1, actual = marketing$sales ) %>%
  ggplot( aes(x = p, y = actual) ) + geom_point() +
  geom_abline(intercept = 0, slope = 1, colour = "red");
```



performance evaluation

```
# Model performance  
data.frame(  
  RMSE = RMSE(predicted.m1, marketing$sales ),  
  R2 = R2(predicted.m1, marketing$sales )  
)
```

	RMSE	R2
1	2.002284	0.8972106

get rid of newspaper

```
m4 <- lm( sales ~ youtube + facebook, data = marketing );
anova(m1, m4);
```

Analysis of Variance Table

Model 1: sales ~ youtube + facebook + newspaper

Model 2: sales ~ youtube + facebook

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	196	801.83				
2	197	801.96	-1	-0.12775	0.0312	0.8599

```
data.frame(
  with_newspapers = summary(m1)$r.squared,
  without_newspapers = summary(m4)$r.squared
)
```

	with_newspapers	without_newspapers
1	0.8972106	0.8971943



relative importance analysis

```
library(relaimpo);
calc.relimp( sales ~ youtube + facebook + newspaper, data = marketing );
```

Response variable: sales

Total response variance: 39.19947

Analysis based on 200 observations

3 Regressors:

youtube facebook newspaper

Proportion of variance explained by model: 89.72%

Metrics are not normalized (rela=FALSE).

Relative importance metrics:

	lmg
youtube	0.58527298
facebook	0.28878652
newspaper	0.02315114

Average coefficients for different model sizes:

interactions

interactions 考虑因素之间的依赖关系或互作关系，比如，在一平台上投放广告会促进另一个平台上广告的效果，因为两个平台的用户可能是重叠的。他们在两个平台都看到广告时，更可能购买产品。

```
m5 <- lm( sales ~ youtube + facebook + youtube:facebook, data = marketing );
anova(m4, m5);
```

Analysis of Variance Table

Model 1: sales ~ youtube + facebook

Model 2: sales ~ youtube + facebook + youtube:facebook

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	197	801.96				
2	196	251.26	1	550.7	429.59	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1



interactions, cont.

```
data.frame(
  no_interactions = summary(m4)$r.squared,
  with_interactions = summary(m5)$r.squared
)
```

	no_interactions	with_interactions
1	0.8971943	0.9677905

结论: 两个平台同时投放广告最好;

interactions, cont.

```
## m5 <- lm( sales ~ youtube + facebook + youtube:facebook, data = marketing );
## 上面的 m5 可以直接写为:
m6 <- lm( sales ~ youtube*facebook, data = marketing );
summary(m6);
```

Call:

```
lm(formula = sales ~ youtube * facebook, data = marketing)
```

Residuals:

Min	1Q	Median	3Q	Max
-7.6039	-0.4833	0.2197	0.7137	1.8295

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.100e+00	2.974e-01	27.233	<2e-16 ***
youtube	1.910e-02	1.504e-03	12.699	<2e-16 ***
facebook	2.886e-02	8.905e-03	3.241	0.0014 **
youtube:facebook	9.054e-04	4.368e-05	20.727	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1



visualize interactions

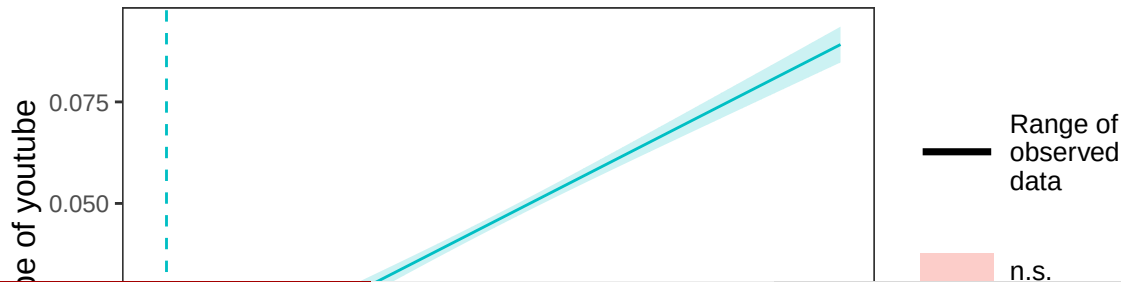
```
library(interactions); ## 装入
sim_slopes(m6, pred = youtube, modx = facebook, jnplot = TRUE)
```

JOHNSON-NEYMAN INTERVAL

When facebook is OUTSIDE the interval $[-26.70, -16.41]$, the slope of youtube is $p < .05$.

Note: The range of observed values of facebook is $[0.00, 59.52]$

Johnson–Neyman plot



relative importance analysis including interactions

```
library(relaimpo);
calc.relimp( sales ~ youtube*facebook, data = marketing );
```

Response variable: sales

Total response variance: 39.19947

Analysis based on 200 observations

3 Regressors:

youtube facebook youtube:facebook

Proportion of variance explained by model: 96.78%

Metrics are not normalized (rela=FALSE).

Relative importance metrics:

	lmg
youtube	0.58851843
facebook	0.30867583
youtube:facebook	0.07059629

Average coefficients for different model sizes:

assumptions of linear regression

重要信息

- ① 任何检验都有基本的假设
- ② 将检验应用于不符合假设的数据是统计学最大的滥用

assumptions

- ① Linearity: The relationship between X and the mean of Y is linear.
- ② Homoscedasticity: The variance of residual is the same for any value of X .
- ③ Independence: Observations are independent of each other.
- ④ Normality: For any fixed value of X , Y is normally distributed.

glm vs. lm

```
lm(formula, data, ...)
```

```
glm(formula, family=gaussian, data, ...)
```

glm:

① 当 family=gaussian 时, 二者是一样的。

```
library(texreg);
m.lm <- lm(am ~ disp + hp, data=mtcars);
m.glm <- glm(am ~ disp + hp, data=mtcars);
screenreg(l = list(m.lm, m.glm))
```

```
=====
              Model 1      Model 2
-----
(Intercept)    0.76 ***    0.76 ***
               (0.16)      (0.16)
disp          -0.00 ***    -0.00 ***
               (0.00)      (0.00)
hp              0.00 *      0.00 *
               (0.00)      (0.00)
```

glm 还可用于其它类型数据的分析

① Logistic regression (family=binomial)

预测的结果 (Y) 是 binary 的分类, 比如 Yes, No, 且只能有两个值;

```
dat <- iris %>% filter( Species %in% c("setosa", "virginica") );
bm <- glm( Species ~ Sepal.Length + Sepal.Width + Petal.Length + Petal.Width,
          data = dat, family = binomial );

data.frame( predicted = bm %>% predict( dat, type = "response" ),
            original = dat$Species ) %>% sample_n(6) %>% arrange( original );
```

	predicted	original
2	4.732528e-13	setosa
44	2.131127e-11	setosa
55	1.000000e+00	virginica
77	1.000000e+00	virginica
75	1.000000e+00	virginica
84	1.000000e+00	virginica

注意:

predict(., type = "response") 的意义是什么??



glm 的 Poisson regression (family=poisson)

Poisson regression is a special type of regression in which the response variable consists of **count data**.

Assumptions:

- ① The response variable consists of count data.
- ② Observations are independent.
- ③ The mean and variance of the model are equal.
- ④ The distribution of counts follows a Poisson distribution.

Section 3

section 3: Non-linear regression (nls)

一元 nls

什么是 nls? to predict a **target variable** using a **non-linear function** consisting of **parameters** and **one or more independent variables**.

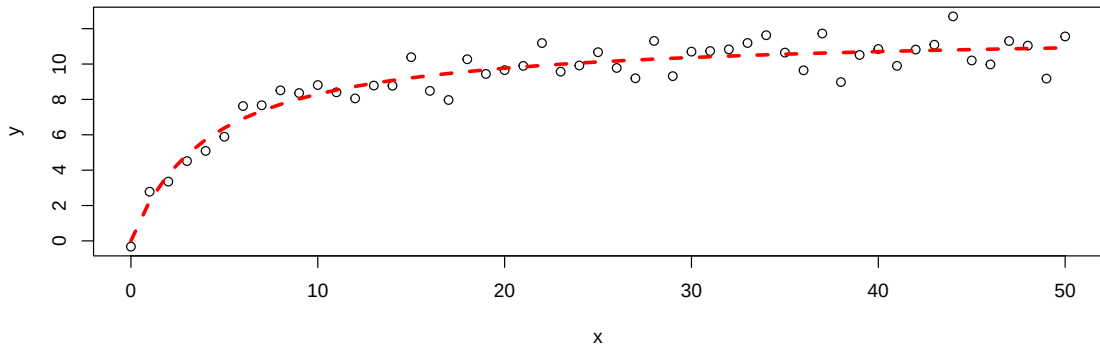
non-linear least squares

给个示例:

```
set.seed(20160227)
x<-seq(0,50,1)
y<-((runif(1,10,20)*x)/(runif(1,0,10)+x))+rnorm(51,0,1)
#for simple models nls find good starting values for the parameters even if it throw a warning
m<-nls(y~a*x/(b+x), start = list(a=0.1, b=0.1));
```

non-linear least squares, cont.

```
#get some estimation of goodness of fit  
plot(x,y)  
lines(x,predict(m),lty=2,col="red",lwd=3)
```



non-linear least squares

```
## 1. generate data
set.seed(20160227)
x<-seq(0,50,1)
y<-((runif(1,10,20)*x)/(runif(1,0,10)+x))+rnorm(51,0,1)

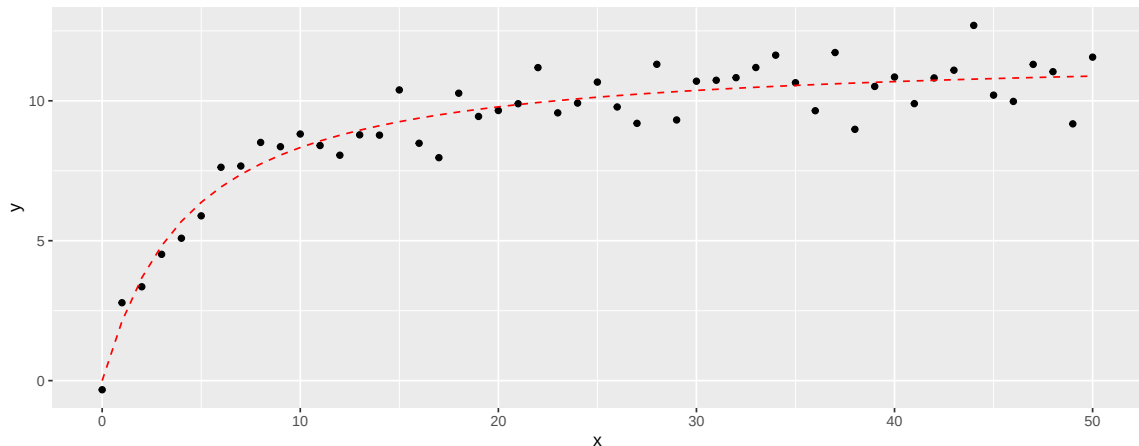
## 2. fit model using nls
m<-nls(y~a*x/(b+x), start = list(a=0.1, b=0.1));

## 3. show how good is the fitting ...
#get some estimation of goodness of fit
plot(x,y)
lines(x,predict(m),lty=2,col="red",lwd=3)
```

- ① `nls( equation, data = data, start = ... )`
- ② `y~a*x/(b+x)`

non-linear least squares, using drc

```
library(drc); ## Analysis of Dose-Response Curves
m13 <- drm( y ~ x, fct = LL.3() ); ## here LL.3() is a least square function
data.frame( x = x, y = y, fitted = predict(m13) ) %>% ggplot( aes(x=x,y=y) ) +
  geom_point() + geom_line( aes(x = x, y = fitted), colour = "red", linetype = 2 );
```



non-linear functions

Polynomials

- Linear equation
- Quadratic polynomial

Concave/Convex curves (no inflection)

- Exponential equation
- Asymptotic equation
- Negative exponential equation
- Power curve equation
- Logarithmic equation
- Rectangular hyperbola

non-linear functions, cont.

Sygmoidal curves

- Logistic equation
- Gompertz equation
- Log-logistic equation (Hill equation)
- Weibull-type 1
- Weibull-type 2

Curves with a maximum

- Brain-Cousens equation

example: Exponential equation

Exponential decay

$$y = a * (\exp(k * X) * k)$$

install packages aomisc and statforbiology:

```
install.packages("statforbiology", "aomisc");
```

```
library(aomisc); library(drc); library(statforbiology); ## updated on Oct 2025
data(degradation);
m14 <- drm(Conc ~ Time, fct = EXD.3(), data = degradation)
summary(m14)
```

Model fitted: Shifted exponential decay (3 parms)

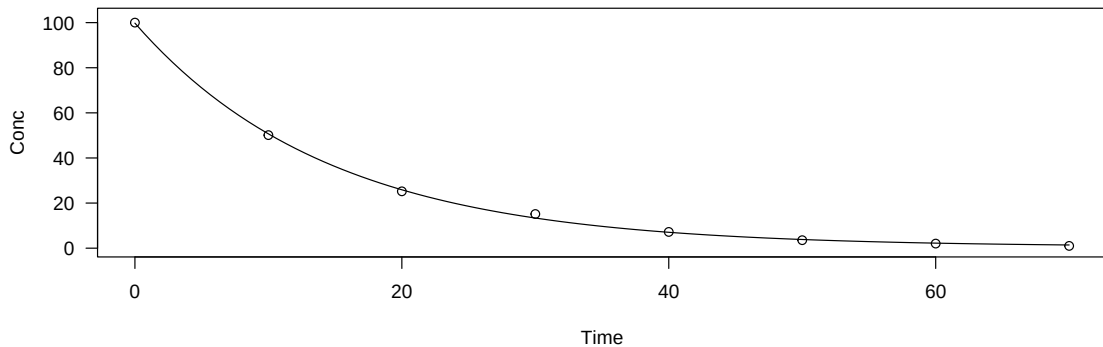
Parameter estimates:

	Estimate	Std. Error	t-value	p-value
c:(Intercept)	0.59264	1.03063	0.575	0.5714
d:(Intercept)	99.77610	1.50721	66.199	< 2.2e-16 ***
e:(Intercept)	14.64011	0.63737	22.970	2.268e-16 ***



plot Exponential decay

```
plot(m14, log="");
```



Power curve

$$a * (X^{(b - 1)} * b)$$

```
data(speciesArea)
m15 <- drm(numSpecies ~ Area, fct = statforbiology::DRC.powerCurve(),
           data = speciesArea)
summary(m15)
```

Model fitted: Power curve (Freundlich equation) (2 parms)

Parameter estimates:

	Estimate	Std. Error	t-value	p-value
a:(Intercept)	4.348404	0.337197	12.896	3.917e-06 ***
b:(Intercept)	0.329770	0.016723	19.719	2.155e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

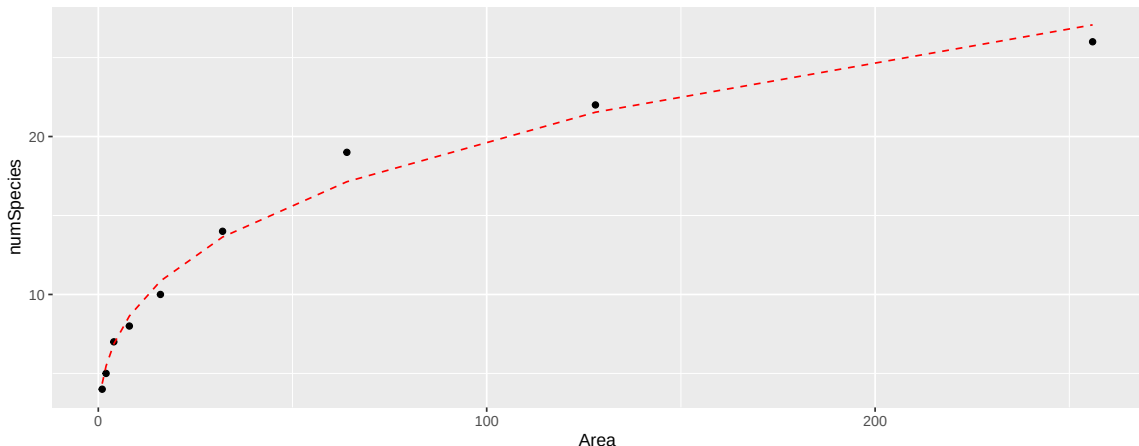
Residual standard error:

0.9588598 (7 degrees of freedom)



Power curve 结果作图

```
speciesArea %>% mutate( fitted = predict( m15, speciesArea ) ) %>%  
  ggplot( aes(x= Area, y= numSpecies) ) +  
  geom_point() + geom_line( aes(x = Area, y = fitted), colour = "red", linetype = 2 );
```



how good is the model?

```
R2( speciesArea$numSpecies, predict( m15, speciesArea ) );
```

```
[1] 0.9874392
```

```
## compare with a linear model
```

```
m16 <- lm( numSpecies ~ Area, data = speciesArea );
```

```
R2( speciesArea$numSpecies, predict( m16, speciesArea ) );
```

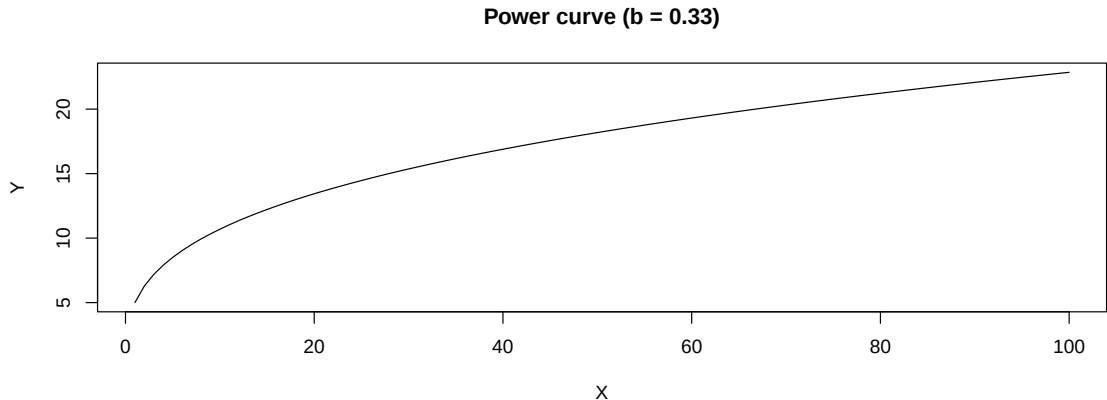
```
[1] 0.8110041
```

数据生成函数

aomisc 包、statforbiology 包和 drc 包带了非常多数据生成的函数，其使用示例如下：

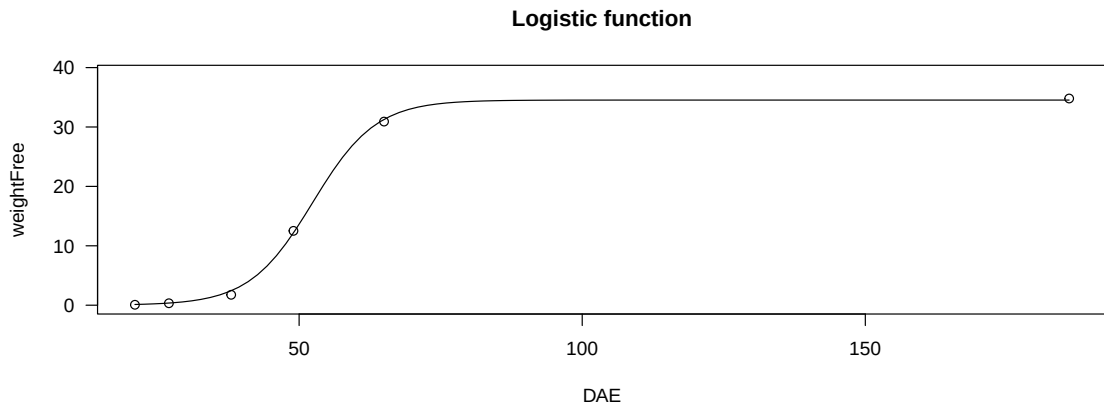
注 $a * (X^{(b - 1)} * b)$ 需要两个参数, a 和 b

```
plot(powerCurve.fun(1:100, 5, 0.33),  
     xlab = "X", ylab = "Y", main = "Power curve (b = 0.33)", type = "l")
```



logistic function

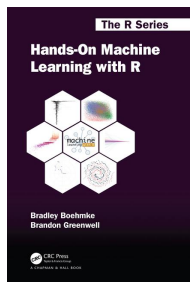
```
data(beetGrowth)
m17 <- drm(weightFree ~ DAE, fct = L.3(), data = beetGrowth)
plot(m17, log="", main = "Logistic function")
```



多元 non-linear regression

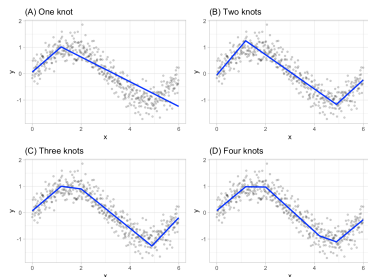
- 1 mars : Multivariate Adaptive Regression Splines (多元自适应回归样条)
- 2 machine learning

推荐一本教程: Hands-On Machine Learning with R



多元 non-linear regression, cont.

Multivariate adaptive regression splines (MARS) provide a convenient approach to capture the nonlinear relationships in the data by assessing cutpoints (knots). The procedure assesses each data point for each predictor as a knot and creates a linear regression model with the candidate feature(s).



fit a basic MARS model, get data ready...

```
library(AmesHousing); ## The Ames Iowa Housing Data
ames <- AmesHousing::make_ames();
head(ames);
```

```
# A tibble: 6 x 81
```

	MS_SubClass <fct>	MS_Zoning <fct>	Lot_Frontage <dbl>	Lot_Area <int>	Street <fct>	Alley <fct>	Lot_Shape <fct>
1	One_Story_1946_and_New~	Resident~	141	31770	Pave	No_A~	Slightly~
2	One_Story_1946_and_New~	Resident~	80	11622	Pave	No_A~	Regular
3	One_Story_1946_and_New~	Resident~	81	14267	Pave	No_A~	Slightly~
4	One_Story_1946_and_New~	Resident~	93	11160	Pave	No_A~	Regular
5	Two_Story_1946_and_New~	Resident~	74	13830	Pave	No_A~	Slightly~
6	Two_Story_1946_and_New~	Resident~	78	9978	Pave	No_A~	Slightly~

```
# i 74 more variables: Land_Contour <fct>, Utilities <fct>, Lot_Config <fct>,
# Land_Slope <fct>, Neighborhood <fct>, Condition_1 <fct>, Condition_2 <fct>,
# Bldg_Type <fct>, House_Style <fct>, Overall_Qual <fct>, Overall_Cond <fct>,
# Year_Built <int>, Year_Remod_Add <int>, Roof_Style <fct>, Roof_Matl <fct>,
# Exterior_1st <fct>, Exterior_2nd <fct>, Mas_Vnr_Type <fct>,
# Mas_Vnr_Area <dbl>, Exter_Qual <fct>, Exter_Cond <fct>, Foundation <fct>,
# Bsmt_Qual <fct>, Bsmt_Cond <fct>, Bsmt_Exposure <fct>, ...
```



fit a basic MARS model, fit a basic model

```
library(earth);
mars1 <- earth( Sale_Price ~ ., data = ames );
print(mars1);
```

Selected 37 of 40 terms, and 26 of 308 predictors

Termination condition: RSq changed by less than 0.001 at 40 terms

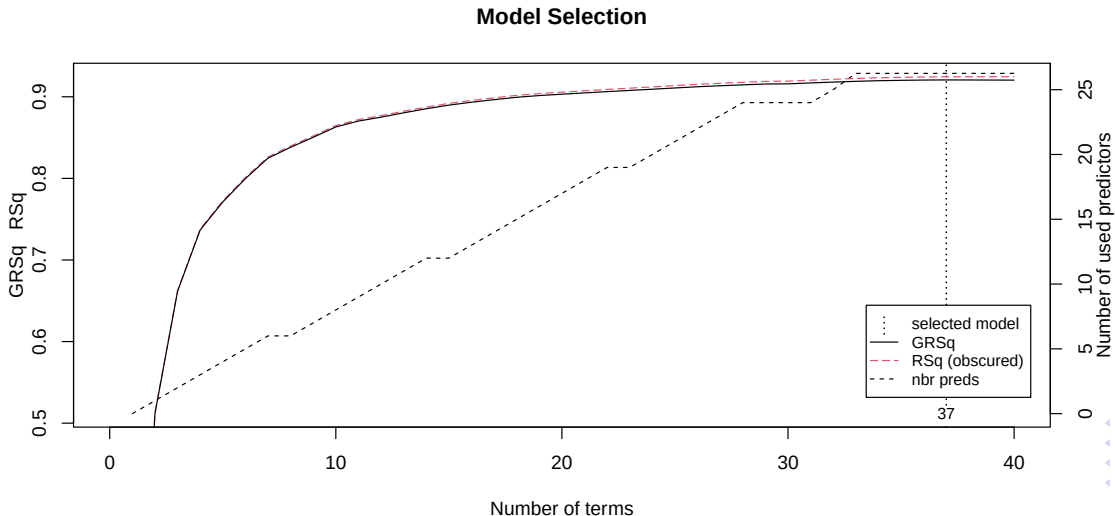
Importance: Gr_Liv_Area, Year_Built, Total_Bsmt_SF, Overall_QualExcellent, ...

Number of terms at each degree of interaction: 1 36 (additive model)

GCV 506531262 RSS 1.411104e+12 GRSq 0.9206569 RSq 0.9245098

MARS model, model selection

```
plot(mars1, which = 1);
```



mars model with interactions

```

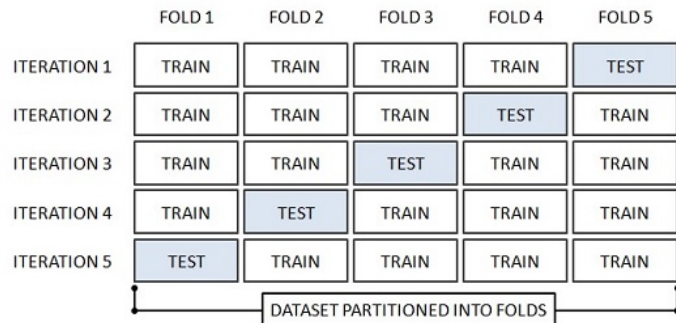
mars2 <- earth(Sale_Price ~ ., data = ames, degree = 2);
summary(mars2) %>% .$coefficients %>% head(10)

```

	Sale_Price
(Intercept)	3.040042e+05
h(Gr_Liv_Area-3228)	1.975129e+02
h(3228-Gr_Liv_Area)	-4.415067e+01
h(Year_Built-2003)	7.976946e+03
h(2003-Year_Built)	-4.970063e+02
h(Total_Bsmt_SF-2452)	4.990177e+01
h(2452-Total_Bsmt_SF)	-5.482326e+01
h(Year_Built-2003)*h(2439-Gr_Liv_Area)	-5.515644e+00
h(2003-Year_Built)*h(Total_Bsmt_SF-1117)	-7.288488e-01
h(2003-Year_Built)*h(1117-Total_Bsmt_SF)	3.682410e-01

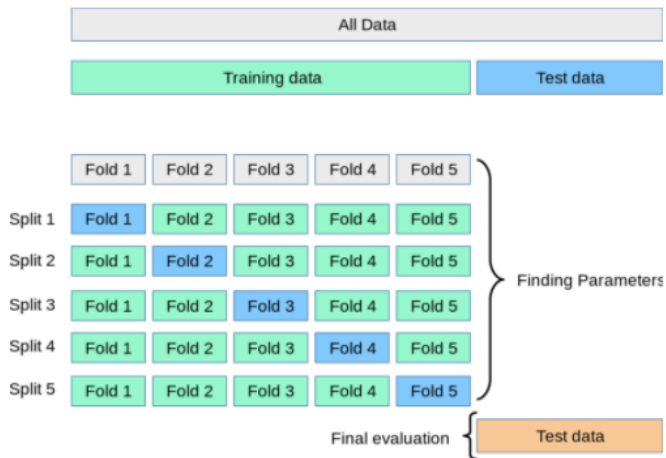
Parameter tuning: cross validation

K fold, N times



注 1. 每个 iteration 为随机 split; 2. 得到 $K * N$ 个模型;

cross validation & additional test



Parameter tuning

```
hyper_grid <- expand.grid(
  degree = 1:3,  ## number of interaction degrees
  nprune = seq(2, 100, length.out = 10) %>% floor() ## number of features to select
)
head(hyper_grid);
```

	degree	nprune
1	1	2
2	2	2
3	3	2
4	1	12
5	2	12
6	3	12

Parameter tuning, cont.

```
set.seed(123) # for reproducibility
cv_mars <- train(
  x = subset(ames, select = -Sale_Price),
  y = ames$Sale_Price,
  method = "earth",
  metric = "RMSE",
  trControl = trainControl(method = "cv", number = 10),
  tuneGrid = hyper_grid
)
```

注 10 折 cross validation

注 非常 time consuming ... 最好能使用 parallel computing ...

Parameter tuning, show results

```
cv_mars$bestTune;
```

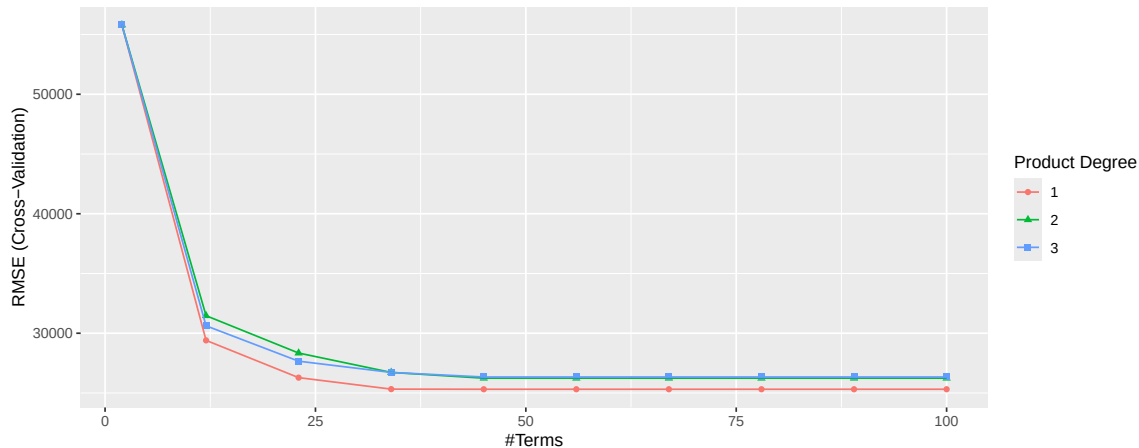
```
  nprune degree
5      45      1
```

```
cv_mars$results %>%
  filter(nprune == cv_mars$bestTune$nprune, degree == cv_mars$bestTune$degree);
```

	degree	nprune	RMSE	Rsquared	MAE	RMSESD	RsquaredSD	MAESD
1	1	45	25312.47	0.8968114	16326.29	4233.975	0.04749767	1172.919

Parameter tuning, plot results, cont.

```
ggplot(cv_mars) ## plot
```



RMSE the lower the better

compare with other methods e.g., lm

```
cv_lm <- train(
  Sale_Price ~ ., data = ames, method = "lm",
  trControl = trainControl(method = "repeatedcv", number = 10, repeats = 3)
);

print(cv_lm);
```

Linear Regression

2930 samples
80 predictor

No pre-processing

Resampling: Cross-Validated (10 fold, repeated 3 times)

Summary of sample sizes: 2637, 2637, 2637, 2637, 2637, 2638, ...

Resampling results:

RMSE	Rsquared	MAE
35829.97	0.8050749	16411.61

Tuning parameter 'intercept' was held constant at a value of TRUE



variable importance plot (VIP)

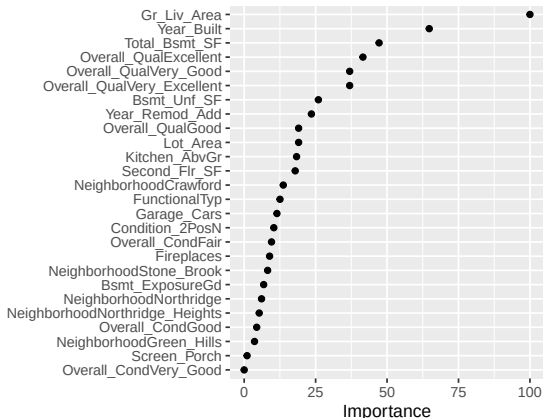
```
library(vip);
```

```
p1 <- vip(cv_mars, num_features = 40, geom = "point", value = "gcv") + ggtitle("Generalized cross-validation")
```

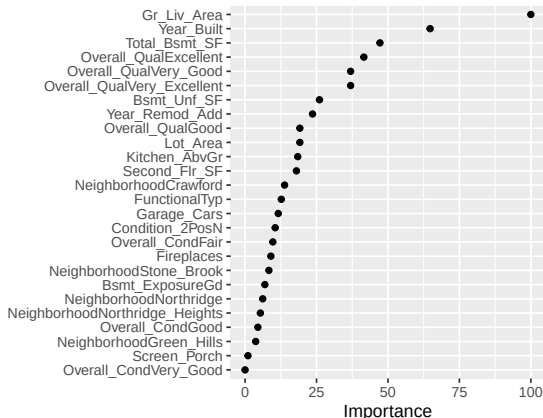
```
p2 <- vip(cv_mars, num_features = 40, geom = "point", value = "rss") + ggtitle("Residual Sums of Squares")
```

```
gridExtra::grid.arrange(p1, p2, ncol = 2)
```

Generalized cross-validation



Residual Sums of Squares



mars : final thoughts

advantages

- ① First, MARS naturally handles mixed types of predictors (quantitative and qualitative).
- ② MARS also requires minimal feature engineering (e.g., feature scaling) and performs automated feature selection.
- ③ Highly correlated predictors do not impede predictive accuracy as much as they do with OLS models.

shortcomings

- ① typically slower to train
- ② Also, although correlated predictors do not necessarily impede model performance, they can make model interpretation difficult.

Section 4

section 4: 小结及作业！

本次小结

linear regression

- lm vs. glm
- 一元
- 多元
- 相关函数
- performance evaluation
- interactions
- visualizations

non-linear regression

- nls
- mars
- earth
- cross validation
- K fold, N times (下次详细讲)

下次预告

- Random Forest
- Support Vector Machine
- Deep learning

作业

- Exercises and homework 目录下 talk11-homework.Rmd 文件;
- 完成时间: 见钉群的要求

important

- all codes are available at Github:
<https://github.com/evolgeniusteam/R-for-bioinformatics>